

Pushdown Automata:-

It is a way to implement context free grammar in a similar way we design finite automata for Regular languages.

2) More powerful than FSM. (Finite State Machine)

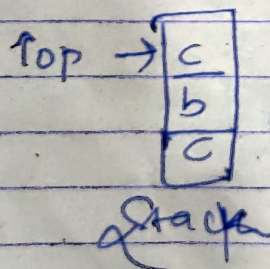
3) ^{FSM} has very limited memory but PDA has more memory.

4) PDA = FSM + stack.

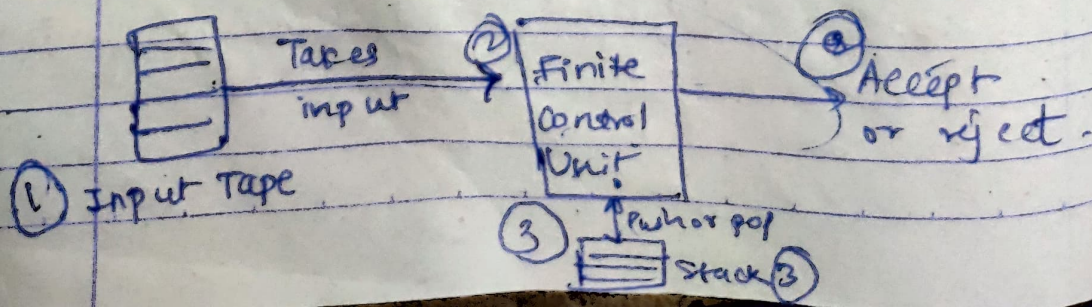
A stack is a way we arrange elements one on top of another.
Two basic operatⁿ

PUSH:- A new element is added at top of stack.

POP:- Top element is removed



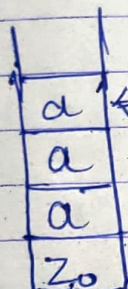
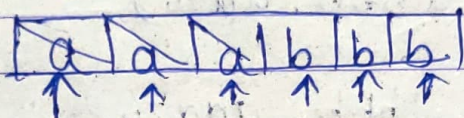
3 components of PDA



Basic eg of PDA

String : $\{a^n b^n / n \geq 1\}$

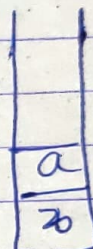
Lang = $\{aaabbb, aaaabbbb\}$



pop, this as b has come



pop this too



pop too



rel 2:-

State:-

state: q_0 - top take
Top stack

$$\delta(q_0, a, Z_0) = \delta(q_0, a, Z_0)$$

$$\delta(q_0, a, a) = \delta(q_0, a, a)$$

$$\delta(q_0, a, a) = \delta(q_0, a, a)$$

No, it pts to b,

$$\text{pop} \rightarrow \delta(q_0, a, a) = \delta(q_0, \epsilon)$$

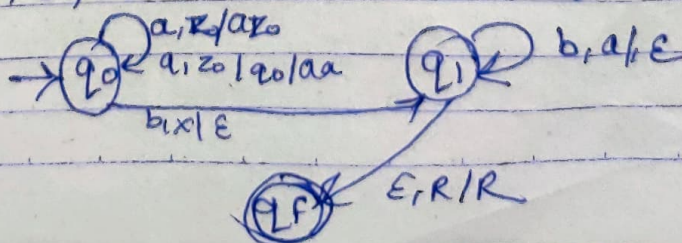
$$\delta(q_0, b, a) = \delta(q_1, \epsilon)$$

$$\delta(q_1, b, a) = \delta(q_1, \epsilon)$$

$$\delta(q_1, b, a) = \delta(q_1, \epsilon)$$

$$\delta(q_1, \epsilon, Z_0) = \delta(q_{LF}, Z_0)$$

Step 3



Step 4:-

$\delta(q_0, aaabbb, \epsilon)$

$\delta(q_0, aabbb, az_0)$

$\delta(q_0, abbb, aaz_0)$

$\delta(q_0, bbb, daaz_0)$

$\delta(q_1, bb, daaz_0)$

$\delta(q_1, b, daaz_0)$

$\delta(q_1, \epsilon, za_0)$

Accepted

$Q = \{q_0, q_1, q_f\}$

$\Sigma = \{a, b\}$

$\Gamma = \{a, b, z_0\}$

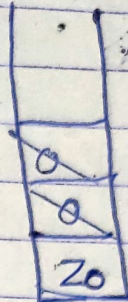
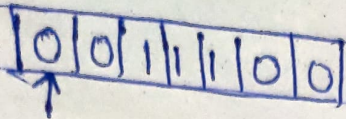
$q_0 =$ initial start state

$z_0 =$ initial stack symbol

$F = \{q_f\}$

$$\{0^n 1^m 0^n \mid m, n \geq 1\}$$

$$L = \{0110, 0011100, 0001111000, \dots\}$$



$$\delta(q_0, 0, z_0) = \delta(q_0, 0, z_0)$$

$$\delta(q_0, 0, 0) = \delta(q_0, 0, 0)$$

$$\delta(q_0, 1, 0) = \delta(q_1, 0)$$

$$\delta(q_1, 1, 0) = \delta(q_1, \epsilon)$$

$$\delta(q_1, 1, 1) = \delta(q_1, 1)$$

$$\delta(q_1, 1, 1)$$

g CFA to PDA

vari

terminal

$$S \rightarrow aAA$$

$$A \rightarrow aslbsla$$

ANF
form

$$(1) \delta(q, a, s) \Rightarrow (q, AA)$$

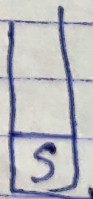
$$(2) \delta(q, a, A) \Rightarrow (q, s)$$

$$(3) \delta(q, b, A) \Rightarrow (q, s)$$

$$(4) \delta(q, a, A) \Rightarrow (q, \epsilon)$$

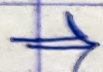
$$S \rightarrow aAA$$

$$aaa$$



~~9aB~~
18 x 16

$S \rightarrow aAA$
 $A \rightarrow as \mid bs \mid a$



✓ Set of terminals = $\{a, b\}$

✓ Set of variables = $\{S, A\}$

$\delta(q, \epsilon, S) = \{(q, aAA)\}$

$\delta(q, \epsilon, A) = \{(q, as), (q, bs), (q, a)\}$

$\delta(q, a, a) = (q, \epsilon)$

$\delta(q, b, b) = (q, \epsilon)$

Q PDA = $\{w \in \Sigma^* \mid w \in \{a, b\}^*$

$\Rightarrow$

$\uparrow \uparrow \uparrow \uparrow \uparrow$
 $\boxed{b \mid a \mid a \mid c \mid a \mid a \mid b \mid}$

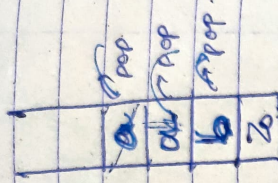
for each $a \rightarrow \text{push } a$

for each $b \rightarrow \text{push } b$

for $c \rightarrow \text{Nope.}$

for each $a \rightarrow \text{pop } a \text{ (stack top} \rightarrow x)$

for each $b \rightarrow \text{pop } b \text{ (stack top} \rightarrow y)$



for Q $\delta(q_0, b, z_0) = \delta(q_0, b, z_0)$

$\delta(q_0, a, b) = \delta(q_0, a, b)$

$\delta(q_0, a, a) = \delta(q_0, a, a)$

for Q .

$\delta(q_0, c, a) = \delta(q_1, a)$

for w

$\delta(q_1, a, a) \rightarrow \delta(q_1, e)$

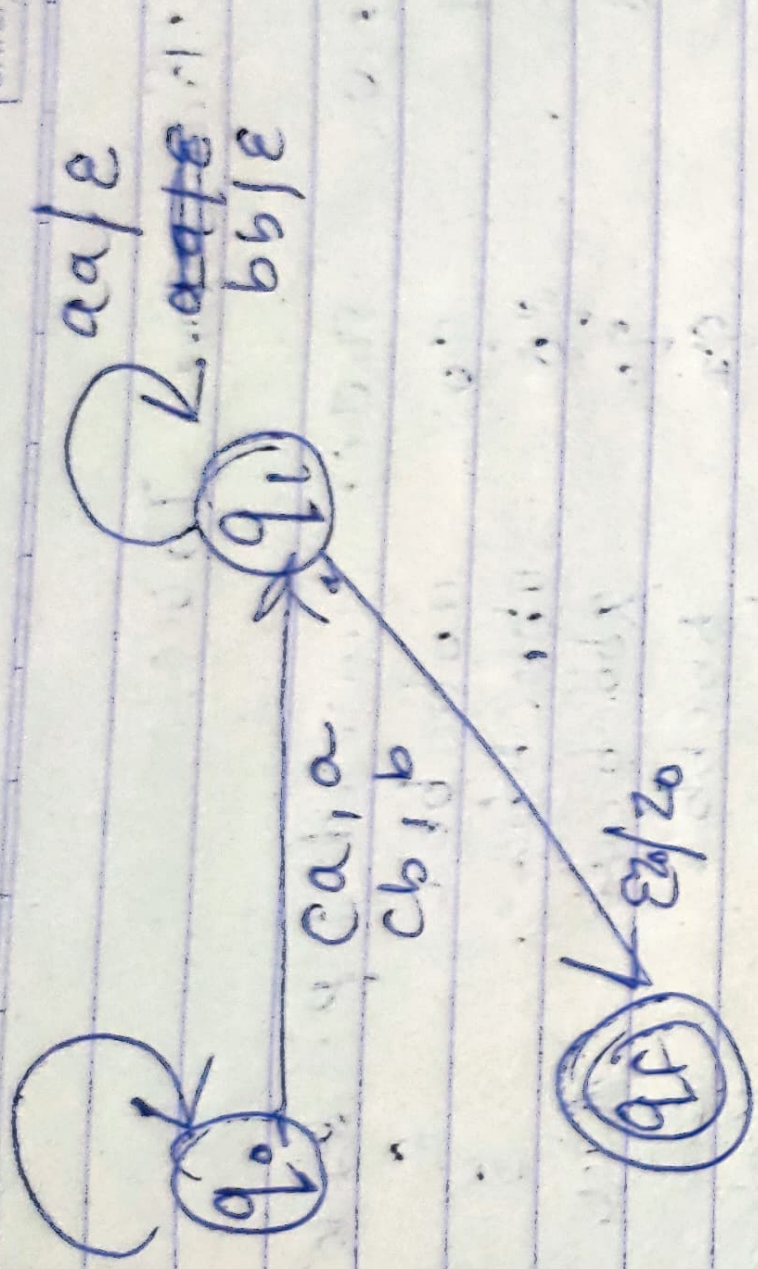
$\delta(q_1, a, a) \rightarrow \delta(q_1, e)$

$\delta(q_1, b, b) \rightarrow \delta(q_1, e)$

for final state.

$(q_1, e, z_0) \Rightarrow (q_f, z_0)$

a, b
 b, q_0, b, z_0



$\delta(q_0, baacab, z_0)$
 $\delta(q_0, aaacab, b, z_0)$